

Personix

Project Team

Tauseef Razaq	21I-1236
Bilal Akbar	21I-1174
Ibraheem Rehman	21I-1102

Session 2021-2025

Supervised by

Dr. Asif Naeem

Co-Supervised by

-



Department of Software Engineering

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

June, 2025

Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	1
1.3	Modules	2
1.3.1	Module 1: Image Preprocessing	2
1.3.2	Module 2: Avatar Creation and Customization	3
1.3.3	Module 3: Animation and Expression	3
1.3.4	Module 4: Content Creation and Sharing	3
1.3.5	Module 5: Cloud Sync and Storage	4
1.3.6	Module 6: User Profile and Settings	4
1.3.7	Module 7: App Development	4
1.4	User Classes and Characteristics	4
2	Project Requirements	6
2.1	Use-case	6
2.1.1	Use Case Diagram	6
2.1.2	High-Level Use Cases	7
2.1.2.1	Use Case 1: Create Avatar	7
2.1.2.2	Use Case 2: Customize Avatar	7
2.1.2.3	Use Case 3: Generate Expressive Video with TTS	7
2.1.2.4	Use Case 4: Generate and Share Content	7
2.1.2.5	Use Case 5: User Registration	8
2.1.3	Fully Dressed Use Cases	8
2.1.3.1	Use Case 1: Create Avatar	8
2.1.3.2	Use Case 2: Customize Avatar	9
2.1.3.3	Use Case 3: Generate Expressive Video with TTS	10
2.1.3.4	Use Case 4: Generate and Share Content	11
2.1.3.5	Use Case 5: User Registration	12
2.2	Functional Requirements	12
2.2.1	Module 1: Avatar Creation and Customization	12
2.2.1.1	Requirements:	13
2.2.2	Module 2: Animation and Expression	13

2.2.2.1	Requirements:	13
2.2.3	Module 3: Content Creation and Sharing	13
2.2.3.1	Requirements:	13
2.2.4	Module 4: Cloud Sync and Storage	14
2.2.4.1	Requirements:	14
2.2.5	Module 5: User Profile and Settings	14
2.2.5.1	Requirements:	14
2.2.6	Module 6: App Development and Deployment	15
2.2.6.1	Requirements:	15
2.3	Non-Functional Requirements	15
2.3.1	Reliability	15
2.3.1.1	Reliability Requirements:	15
2.3.2	Usability	15
2.3.2.1	Usability Requirements:	16
2.3.3	Performance	16
2.3.3.1	Performance Requirements:	16
2.3.4	Security	16
2.3.4.1	Security Requirements:	16
2.4	Domain Model	17
3	System Overview	18
3.1	Architectural Design	18
3.1.1	Box-and-Line Diagram	18
3.1.2	Modular Program Structure	19
3.1.3	Architecture Style	20
3.1.4	Detailed Mapping of Components to Architecture	21
3.2	Design Models	21
3.2.1	Activity Diagram	21
3.2.1.1	Description:	21
3.2.1.2	Diagram:	22
3.2.2	Class Diagram	28
3.2.2.1	Diagram:	28
3.2.3	Sequence Diagram	28
3.2.3.1	Use Case: User Registration	28
3.2.3.2	Use Case: Avatar Creation	29
3.2.3.3	Use Case: Avatar Customization	30
3.2.3.4	Use Case: Generate Expressive Video with TTS	31
3.2.3.5	Use Case: Generate and Share Content	32
3.2.4	State Transition Diagram	33
3.2.4.1	Description:	33

CONTENTS

3.2.4.2	Diagram:	34
3.3	Data Design	34
References		36

List of Figures

2.1	Use Case Diagram	6
2.2	Domain model	17
3.1	Box and line diagram	19
3.2	Activity diagram: Avatar Creation	22
3.3	Activity Diagram: Avatar Customization	23
3.4	Activity Diagram: Generate Expressive Video with TTS	24
3.5	Use Case 4: Generate and Share Content	26
3.6	Activity Diagram: User Registration	27
3.7	Enter Caption	28
3.8	User Registration	29
3.9	Create Avatar	30
3.10	Avatar Customization	31
3.11	Generate Expressive Video with TTS	32
3.12	Generate and share Content	33
3.13	state machine diagram	34
3.14	ERD-Diagram	35

List of Tables

2.1	High-Level Use Case: Create Avatar	7
2.2	High-Level Use Case: Customize Avatar	7
2.3	High-Level Use Case: Generate Expressive Video with TTS	7
2.4	High-Level Use Case: Generate and Share Content	7
2.5	High-Level Use Case: User Registration	8
2.6	Fully Dressed Use Case: Create Avatar	8
2.7	Fully Dressed Use Case: Customize Avatar	9
2.8	Fully Dressed Use Case: Generate Expressive Video with TTS	10
2.9	Fully Dressed Use Case: Generate and Share Content	11
2.10	Fully Dressed Use Case: User Registration	12

Chapter 1

Introduction

This report outlines the development progress of the Personix platform, which allows users to create personalized 3D avatars from photos, customize their facial features, and animate them with natural expressions. The platform is designed for users who want to create and share personalized avatars and media.

The report is aimed at the evaluation panel, our project supervisor, and for future reference in Software Requirements Specification (SRS) documents during our careers. It provides a clear picture of the project's purpose, its current status, and the next steps in development.

1.1 Problem Statement

In today's digital world, many individuals are hesitant to share their real faces on social media due to privacy concerns or lack of confidence. Existing avatar solutions don't offer enough customization or natural, realistic expressions, limiting users from fully engaging with visual content. Additionally, creating videos for social media can be time-consuming and stressful, often requiring multiple retakes and ideal conditions.

Personix addresses these issues by offering users the ability to create highly customizable, realistic 3D avatars that can express emotions naturally. This helps users confidently participate in online spaces while protecting their privacy.

1.2 Scope

The scope of the Personix project involves developing an Android application that enables users to create realistic 3D avatars using their own photos. The application will use advanced computer vision and facial recognition algorithms to map key facial features,

ensuring that the transition from 2D images to 3D models is accurate and lifelike. Users will have access to a wide range of customization options to modify their avatars, such as adjusting skin tone, hairstyles, facial structure, and facial hair. These customization tools ensure that users can create avatars that closely reflect their personal appearance or their desired digital representation.

A core feature of Personix is the ability to animate avatars with natural facial expressions and movements. The platform will also integrate text-to-speech functionality, allowing avatars to speak in a realistic manner. Sentiment analysis will further enhance the experience by enabling avatars to display appropriate emotions based on the text input provided by the user. Users will be able to generate content such as stickers, GIFs, and short videos (30 seconds to 1 minute), which can easily be shared across various social media platforms.

In addition to creating a single avatar, the platform will support the creation and management of multiple avatars, allowing users to switch between different representations as needed. Users will also have the option to apply their avatars as face filters for video recordings, making the platform even more interactive. Cloud storage functionality will ensure that avatars are securely stored and synchronized across devices, providing users with the flexibility to access their avatars from anywhere. In summary, the scope of Personix includes 3D modeling, avatar customization, animation, text-to-speech integration, content creation tools, and cloud synchronization, all designed to enhance digital self-representation while maintaining user privacy and flexibility.

1.3 Modules

1.3.1 Module 1: Image Preprocessing

This module is responsible for verifying and preparing user-uploaded images for 3D avatar generation. It ensures that the provided images meet the requirements for accurate model creation by analyzing whether they display the necessary angles of the face.

1. Takes image from user
2. Processes the image
3. Displays message based on result.

1.3.2 Module 2: Avatar Creation and Customization

This module focuses on creating realistic 3D avatars from user photos. It allows users to customize facial features, skin tone, and hairstyles, enabling personalized and unique avatar creation. Users can also create and manage multiple avatars and save their presets for later use.

1. 3D model generation from user photos
2. Facial feature customization (skin tone, hairstyle, etc.)
3. Multiple avatar creation and management
4. Save and load avatar

1.3.3 Module 3: Animation and Expression

This module manages the animation of avatars.. It integrates text-to-speech functionality, allowing avatars to speak while displaying emotions based on the sentiment of the text.

1. Expression customization library
2. Text-to-speech integration with sentiment analysis
3. Voice customization options

1.3.4 Module 4: Content Creation and Sharing

This module provides tools for generating shareable content such as stickers, GIFs, and short videos. Users can add avatars to face filters for videos and easily share the content on social media.

1. Sticker creation tools
2. GIF maker with avatar integration
3. Video recording with avatar face filters
4. In-app sharing to social media

1.3.5 Module 5: Cloud Sync and Storage

This module ensures that user avatars and content are securely stored in the cloud and synchronized across multiple devices. It includes tools for managing cloud storage and accessing avatars from any device.

1. Secure cloud storage for avatars and content
2. Cross-device synchronization
3. Cloud storage management tools

1.3.6 Module 6: User Profile and Settings

This module manages user accounts, app preferences, and settings. It allows users to register, customize their profiles, manage notifications, and adjust app settings according to their preferences.

1. User registration and authentication
2. Profile customization
3. Notification management
4. App settings and preferences

1.3.7 Module 7: App Development

This module covers the overall development of the app, including design, functionality, and performance optimization. It also involves deploying the app to app stores and providing updates.

1. User interface (UI) and user experience (UX) design
2. Core functionality implementation
3. Performance optimization and testing
4. App store deployment and updates

1.4 User Classes and Characteristics

Following table shows the user classes along with their description.

User class	Description
End User (Avatar Creator)	The End User is a person who wants to create customized 3D avatars for personal or social media use. They will upload photos, customize avatars, and generate media assets such as GIFs, stickers, and videos. This class includes a diverse range of individuals who value privacy and creativity, typically interacting with the app on a regular basis.
Content Creator	Content Creators are advanced users who want to generate high-quality, shareable content using their avatars. They may frequently use features like video recording with face filters and sharing tools. Many of these users are active on social media platforms and want engaging, customizable avatars for their posts.
Casual User	A Casual User interacts with the app infrequently and may only use basic avatar customization features without diving deep into advanced animation or content creation. They may use the app for fun or one-time avatar creation without saving or frequently using the avatars.
Tester/QA	Testers or Quality Assurance (QA) personnel are responsible for testing the app's features, ensuring that avatar creation, customization, and content generation functionalities work as expected. They focus on identifying bugs, issues, or inconsistencies across different devices.
Developer	Developers are part of the team responsible for the technical development of the application. They work on updating the features, fixing bugs, improving performance, and adding new functionality as required. They are responsible for ensuring the app functions well across various devices and platforms.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case

This section outlines the primary use cases for the Personix application, focusing on how users will interact with the platform to create and customize 3D avatars, animate them, and generate media content.

2.1.1 Use Case Diagram

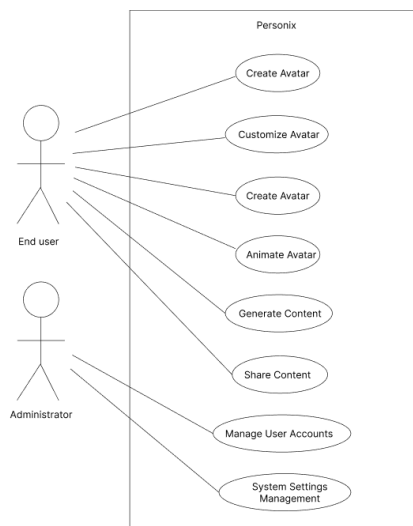


Figure 2.1: Use Case Diagram

2.1.2 High-Level Use Cases

2.1.2.1 Use Case 1: Create Avatar

Use Case Name	Create Avatar
Actors	End User
Type	Primary
Description	The user uploads a photo, and the system generates a 3D avatar based on facial features.

Table 2.1: High-Level Use Case: Create Avatar

2.1.2.2 Use Case 2: Customize Avatar

Use Case Name	Customize Avatar
Actors	End User
Type	Primary
Description	The user customizes the 3D avatar's facial features, hair, and other facial elements.

Table 2.2: High-Level Use Case: Customize Avatar

2.1.2.3 Use Case 3: Generate Expressive Video with TTS

Use Case Name	Generate Expressive Video with TTS
Actors	End User
Type	Primary
Description	The user enters a script and selects their 3D avatar. The system generates a video where the avatar speaks the script using TTS and animates expressions based on sentiment analysis.

Table 2.3: High-Level Use Case: Generate Expressive Video with TTS

2.1.2.4 Use Case 4: Generate and Share Content

Use Case Name	Generate and Share Content
Actors	End User
Type	Primary
Description	The user generates content (e.g., stickers, GIFs) and shares it on social media.

Table 2.4: High-Level Use Case: Generate and Share Content

2.1.2.5 Use Case 5: User Registration

Use Case Name	User Registration
Actors	End User
Type	Primary
Description	The user registers with the system by providing required details.

Table 2.5: High-Level Use Case: User Registration

2.1.3 Fully Dressed Use Cases

2.1.3.1 Use Case 1: Create Avatar

Use Case Name	Create Avatar
Scope	Personix Platform
Level	User Goal
Primary Actor	End User
Stakeholders and Interests	- End User: Wants to create an avatar. - Platform: Ensures proper avatar generation from photos.
Preconditions	- User has uploaded a valid photo. - User is logged into their account.
Postconditions	- A 3D avatar is generated. - The avatar is available for customization.
Main Success Scenario	1. User uploads a valid photo. 2. The system processes the photo using facial recognition. 3. The system generates a 3D avatar.
Extensions	2a. Invalid photo: - The system prompts the user to upload a valid photo. 3a. Avatar generation error: - The system alerts the user and retries the avatar generation.

Table 2.6: Fully Dressed Use Case: Create Avatar

2.1.3.2 Use Case 2: Customize Avatar

Use Case Name	Customize Avatar
Scope	Personix Platform
Level	User Goal
Primary Actor	End User
Stakeholders and Interests	- End User: Wants to personalize their avatar. - Platform: Offers a variety of customization options.
Preconditions	- User has a generated 3D avatar. - User is logged in.
Postconditions	- Avatar is customized. - Customizations are saved for future use.
Main Success Scenario	1. User selects an avatar feature (e.g., skin tone). 2. The system displays customization options. 3. User confirms changes. 4. The system updates the avatar with the new features.
Extensions	3a. Invalid customization selection: - The system prompts the user to select valid customization options. 4a. Customization error: - The system retries the customization process.

Table 2.7: Fully Dressed Use Case: Customize Avatar

2.1.3.3 Use Case 3: Generate Expressive Video with TTS

Use Case Name	Generate Expressive Video with TTS
Scope	Personix Platform
Level	User Goal
Primary Actor	End User
Stakeholders and Interests	- End User: Wants to create videos using their 3D avatars. - Platform: Processes TTS and sentiment analysis to animate avatars.
Preconditions	- User has a customized 3D avatar. - User provides a valid script.
Postconditions	- A video is generated with the avatar speaking the script. - The avatar's expressions match the sentiment analysis of the text.
Main Success Scenario	1. User selects their 3D avatar. 2. User enters a script. 3. The system applies sentiment analysis to the script. 4. The system animates the avatar's expressions. 5. The system processes the script using TTS. 6. The system generates a video with the avatar speaking the text.
Extensions	3a. Invalid script format: - The system prompts the user to enter a valid script. 5a. TTS failure: - The system notifies the user and retries the TTS processing. 6a. Video generation error: - The system alerts the user and retries generating the video.

Table 2.8: Fully Dressed Use Case: Generate Expressive Video with TTS

2.1.3.4 Use Case 4: Generate and Share Content

Use Case Name	Generate and Share Content
Scope	Personix Platform
Level	User Goal
Primary Actor	End User
Stakeholders and Interests	- End User: Wants to create and share personalized content. - Platform: Facilitates content creation and sharing.
Preconditions	- User has customized their avatar. - User is logged in.
Postconditions	- Content is generated and available for sharing. - The content is shared on selected platforms.
Main Success Scenario	1. User selects content type (e.g., video, GIF). 2. The system generates the content. 3. The user shares the content on social media.
Extensions	2a. Content generation failure: - The system retries content generation or notifies the user. 3a. Social media sharing error: - The system retries sharing or prompts the user to try again.

Table 2.9: Fully Dressed Use Case: Generate and Share Content

2.1.3.5 Use Case 5: User Registration

Use Case Name	User Registration
Scope	Personix Platform
Level	User Goal
Primary Actor	End User
Stakeholders and Interests	- End User: Wants to register and access the platform. - Platform: Ensures secure registration and account management.
Preconditions	- User is not already registered. - Registration form is accessible to new users.
Postconditions	- A new user account is created. - User is redirected to the login page.
Main Success Scenario	1. User navigates to the registration page. 2. User fills out the registration form (name, email, password). 3. The system validates the information provided. 4. The system creates a new user account. 5. The system redirects the user to the login page.
Extensions	3a. Invalid email or weak password: - The system prompts the user to correct the input. 4a. Account creation failure: - The system notifies the user and retries the registration. 5a. Redirection failure: - The system logs the user into their newly created account directly.

Table 2.10: Fully Dressed Use Case: User Registration

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

2.2.1 Module 1: Avatar Creation and Customization

This module focuses on creating and customizing 3D avatars. It covers generating avatars from user photos and allowing the user to customize facial features.

2.2.1.1 Requirements:

1. **FR1:** The system shall allow users to upload photos for 3D avatar generation.
2. **FR2:** The system shall generate a 3D avatar based on the uploaded photo.
3. **FR3:** The system shall provide tools for facial feature customization (e.g., hairstyle, facial hairs, facial features).
4. **FR4:** The system shall allow users to save customized avatar for future use.
5. **FR5:** The system shall support creating multiple avatars per user.

2.2.2 Module 2: Animation and Expression

This module handles real-time avatar animation, including facial movements and voice interaction using text-to-speech.

2.2.2.1 Requirements:

1. **FR1:** The system shall allow real-time tracking of user facial movements and map them to the 3D avatar.
2. **FR2:** The system shall support text-to-speech conversion with avatar lip synchronization.
3. **FR3:** The system shall provide sentiment analysis to adjust facial expressions based on the tone of the speech.
4. **FR4:** The system shall allow voice customization, including pitch and tone adjustments.

2.2.3 Module 3: Content Creation and Sharing

This module enables users to create content, such as stickers, GIFs, and videos, and share them through social media.

2.2.3.1 Requirements:

1. **FR1:** The system shall provide tools for users to create custom stickers using their avatars.

2. **FR2:** The system shall allow users to generate GIFs with avatar animations.
3. **FR3:** The system shall support recording videos with avatar face filters.
4. **FR4:** The system shall provide integration with social media platforms for sharing created content.
5. **FR5:** The system shall allow in-app sharing of content with other users.

2.2.4 Module 4: Cloud Sync and Storage

This module focuses on storing avatars and content in the cloud and synchronizing them across devices.

2.2.4.1 Requirements:

1. **FR1:** The system shall securely store user avatars and content in the cloud.
2. **FR2:** The system shall provide synchronization of avatars and content across multiple devices.
3. **FR3:** The system shall allow users to manage cloud storage, including deleting avatars.
4. **FR4:** The system shall support automatic backup of avatars.

2.2.5 Module 5: User Profile and Settings

This module manages user profiles, authentication, and application settings.

2.2.5.1 Requirements:

1. **FR1:** The system shall support user registration and authentication.
2. **FR2:** The system shall allow users to customize their profiles, including avatar preferences.
3. **FR3:** The system shall enable users to configure application preferences such as notifications and privacy settings.
4. **FR4:** The system shall allow users to manage notifications that includes enabling or disabling email alerts and app notification.

2.2.6 Module 6: App Development and Deployment

This module covers the technical aspects of the app development process, including UI/UX, functionality, and optimization.

2.2.6.1 Requirements:

1. The system shall provide a user-friendly and responsive interface for both mobile and web platforms.
2. The system shall implement core features defined in the use cases.
3. The system shall support app deployment and updates through android play store.

2.3 Non-Functional Requirements

2.3.1 Reliability

The system should ensure high reliability to prevent failure during critical operations such as avatar creation and content sharing.

2.3.1.1 Reliability Requirements:

1. The system shall achieve a mean time between failures (MTBF) of at least 1,000 hours during avatar creation processes.
2. System failures during content sharing shall not result in data loss and should automatically retry the operation.
3. The system shall implement error detection mechanisms to log any failures and alert the user with corrective suggestions.
4. In case of failure during avatar creation, the system shall save the avatar's progress and allow the user to resume from the last saved state.

2.3.2 Usability

The system should be user-friendly, easy to navigate, and require minimal training for both new and returning users.

2.3.2.1 Usability Requirements:

1. The system shall allow a user to create and customize an avatar with no more than 3 interactions.
2. The system shall provide tooltip guidance for all customizable features, helping users understand each option.
3. The system shall allow users to recover from any customization mistakes with a single undo action.
4. The system shall be optimized for performance, ensuring smooth operation on devices with limited resources.

2.3.3 Performance

The system should maintain efficient performance across different modules and operations.

2.3.3.1 Performance Requirements:

1. 90% of avatar customization actions shall be reflected on-screen within 3 seconds of the user's interaction.
2. All content-sharing operations shall complete within 10 seconds over a 10 Mbps connection.
3. The system shall handle up to 100 concurrent users customizing avatars without performance degradation.

2.3.4 Security

The system should protect user data, including personal information, and prevent unauthorized access.

2.3.4.1 Security Requirements:

1. All user data, including avatar customizations and personal details, shall be encrypted using AES-256 during storage and transmission.
2. The system shall implement a two-factor authentication (2FA) mechanism for accessing user accounts.

2.4 Domain Model

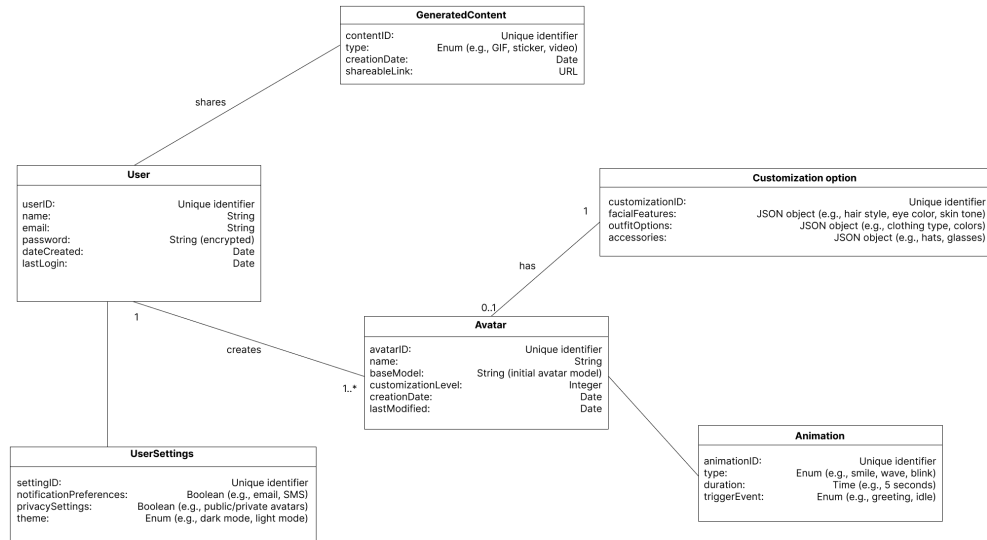


Figure 2.2: Domain model

Chapter 3

System Overview

Personix is an Android application focused on improving digital self-representation through 3D facial models. Users can upload photographs, which are processed using computer vision and facial recognition algorithms to generate detailed 3D avatars. These avatars can be customized to match the user's preferences, allowing changes to features such as head and facial hair, and facial structure. Additionally, Personix integrates text-to-speech technology with sentiment analysis, enabling the avatars to speak and exhibit appropriate facial expressions. The platform also supports media creation, including GIFs and videos, and offers cloud storage for cross-device synchronization.

3.1 Architectural Design

This section provides a high-level overview of the Architectural Design for the Personix's system. The system is decomposed into several key modules that collaborate to achieve the desired core functionality of Personix. The relationships between the modules are also explained to show how data flows between them to ensure seamless operation.

3.1.1 Box-and-Line Diagram

Below is the initial box-and-line diagram representing the major modules of the system and how they interact with each other:

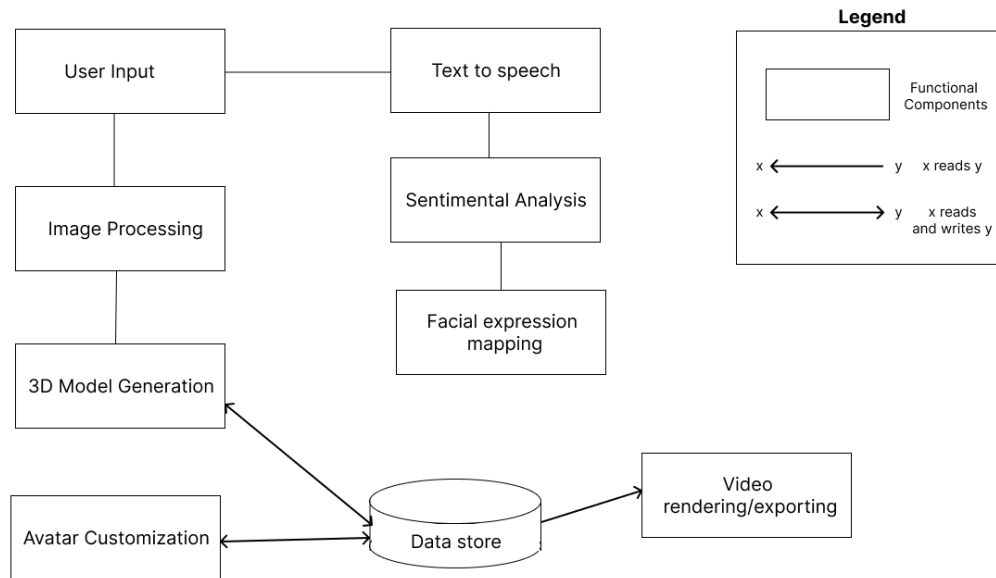


Figure 3.1: Box and line diagram

3.1.2 Modular Program Structure

The system is designed in a modular manner, where each module plays a distinct role in achieving the overall functionality. Below is a brief explanation of the major modules and their relationships:

- **User Input:** Collects user data such as images and text, which is passed to other processing components.
- **Image Processing:** Processes the image input for validation and then use in 3D Model Generation.
- **3D Model Generation:** Based on the processed images, this module generates a 3D avatar. The generated model is stored in the **Data Store** for further use or customization.
- **Text to Speech:** Converts user-provided text into speech and passes the resulting speech to the **Sentiment Analysis** module.
- **Sentiment Analysis:** Analyzes the emotional tone of the speech input and sends the results to **Facial Expression Mapping**.
- **Facial Expression Mapping:** Maps the corresponding facial expressions onto the 3D avatar based on the sentiment analysis.

- **Avatar Customization:** Allows users to modify and customize their 3D avatars. These modifications are stored back in the **Data Store**.
- **Data Store:** Centralized storage for the 3D models and other data. It is accessed by both **Avatar Customization** and **Video Rendering/Exporting**.
- **Video Rendering/Exporting:** Generates videos based on the customized avatars and provides options for exporting or sharing the content.

Some of the modules operate independently, but together they contribute to the system's goal of creating, customizing, and rendering expressive avatars.

3.1.3 Architecture Style

The system adopts a **Layered Architecture**, separating concerns into distinct layers to enhance maintainability and scalability. Below is a description of each layer:

- **Presentation Layer:** This layer consists of modules responsible for user interaction and data presentation. Modules such as:
 - **User Input**
 - **Avatar Customization**
 - **Video Rendering/Exporting**

belong here, as they handle user-facing functionality.

- **Business Logic Layer:** Contains the core functional modules that perform processing tasks, such as:
 - **Image Processing**
 - **3D Model Generation**
 - **Text to Speech**
 - **Sentiment Analysis**
 - **Facial Expression Mapping**

fall under this layer, where the main processing of user data occurs.

- **Data Layer:** This layer is responsible for data storage and retrieval. The:
 - **Data Store**

module resides here, as it manages the persistent storage of avatar models and related data.

3.1.4 Detailed Mapping of Components to Architecture

Here's a detailed mapping of the modules from the box-and-line diagram to the architecture layers:

- **Presentation Layer:**
 - User Input
 - Avatar Customization
 - Video Rendering/Exporting
- **Business Logic Layer:**
 - Image Processing
 - 3D Model Generation
 - Text to Speech
 - Sentiment Analysis
 - Facial Expression Mapping
- **Data Layer:**
 - Data Store

3.2 Design Models

This section provides detailed design models for the system based on an object-oriented development approach. Each model is explained in detail, illustrating how different components of the system interact and function.

3.2.1 Activity Diagram

3.2.1.1 Description:

The activity diagram represents the dynamic aspects of the system by showing the flow of activities and actions between different components. It is used to model the workflow of the system.

- **Purpose:** The activity diagram helps visualize the flow of tasks, actions, and decisions that a user or the system performs to achieve a specific functionality.
- **Components:** User Input, 3D Avatar Customization, Text-to-Speech, Sentiment Analysis, Video Export, etc.

3.2.1.2 Diagram:

Use Case 1: Avatar Creation

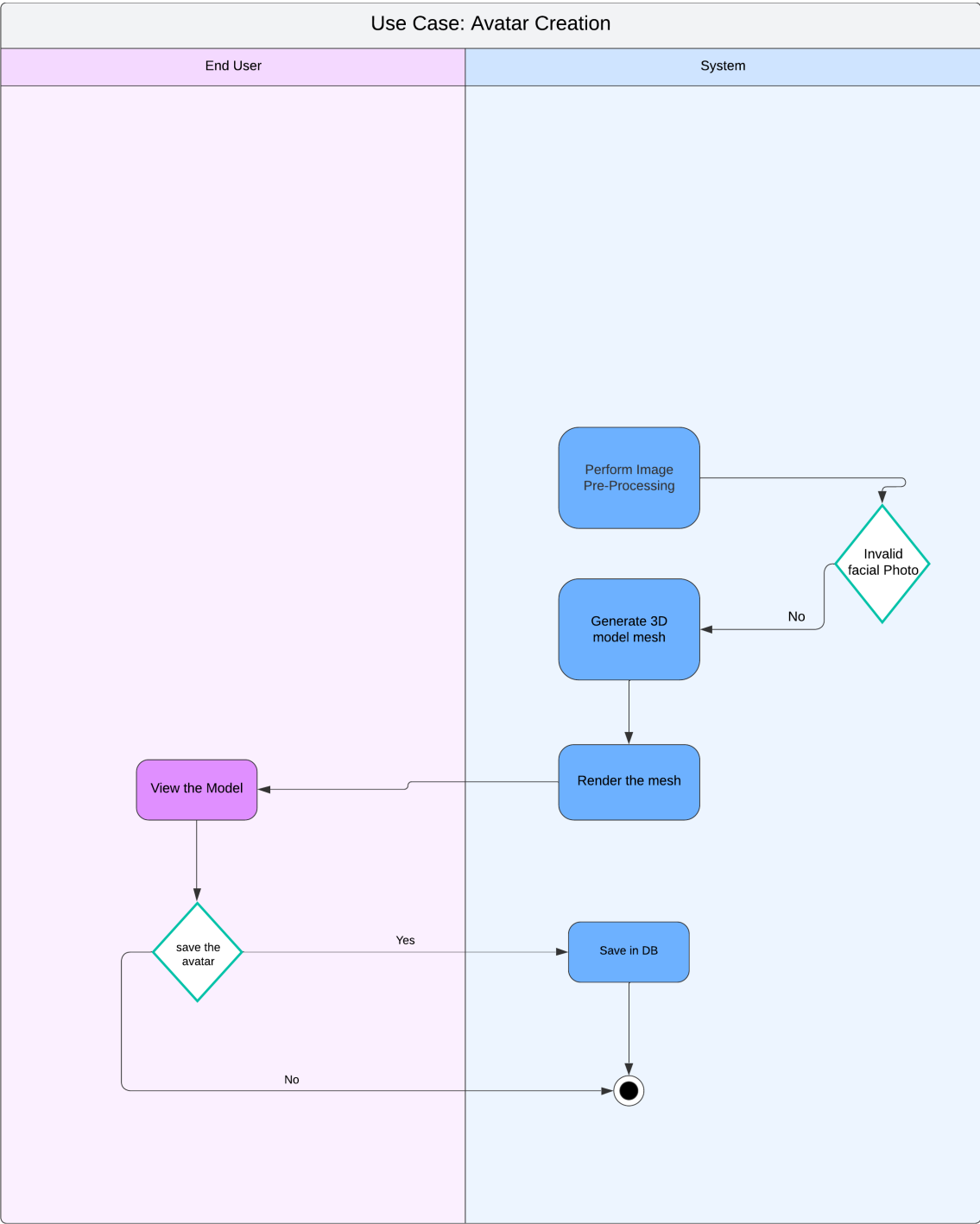


Figure 3.2: Activity diagram: Avatar Creation

Use Case 2: Avatar Customization

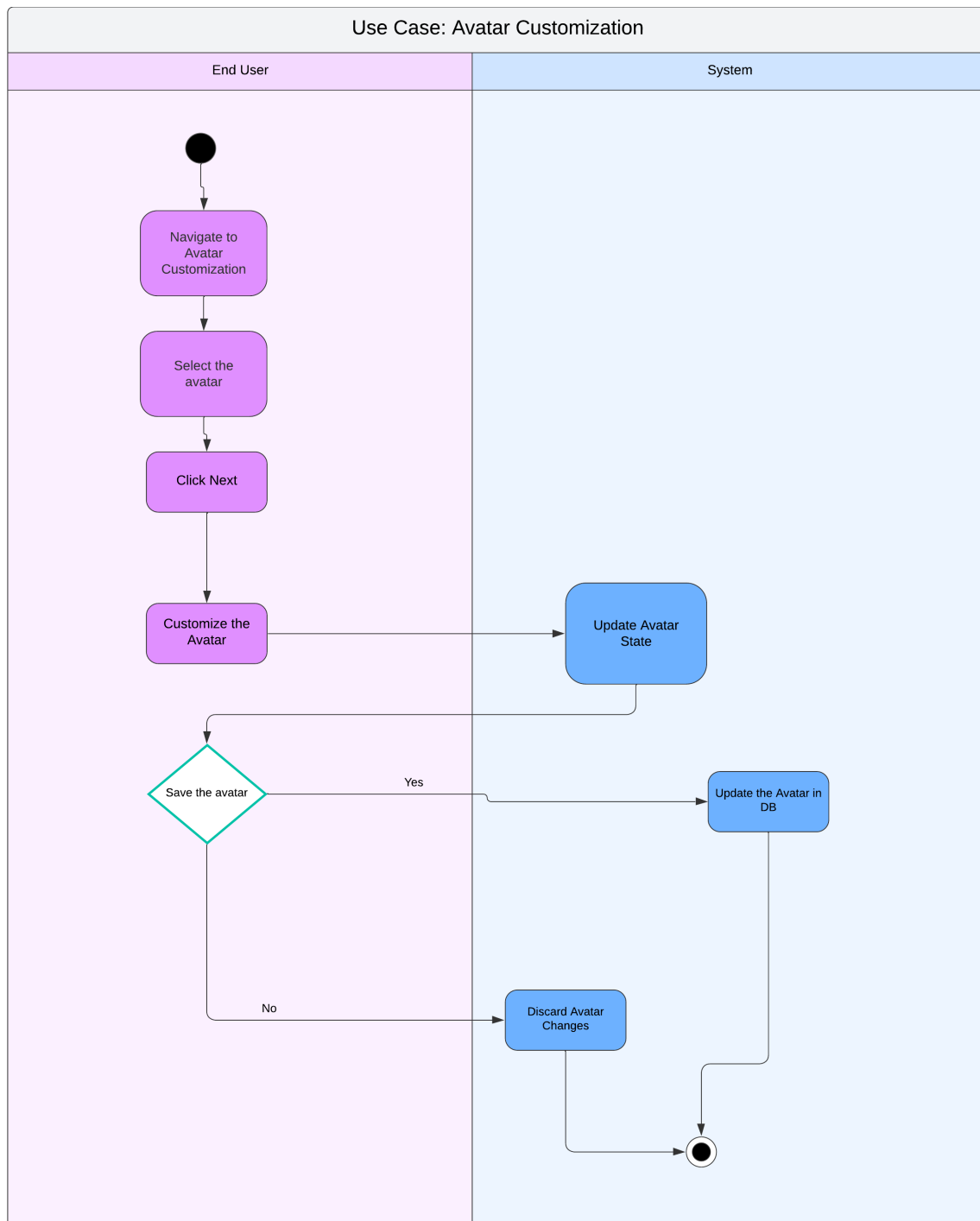


Figure 3.3: Activity Diagram: Avatar Customization

Use Case 3: Generate Expressive Video with TTS

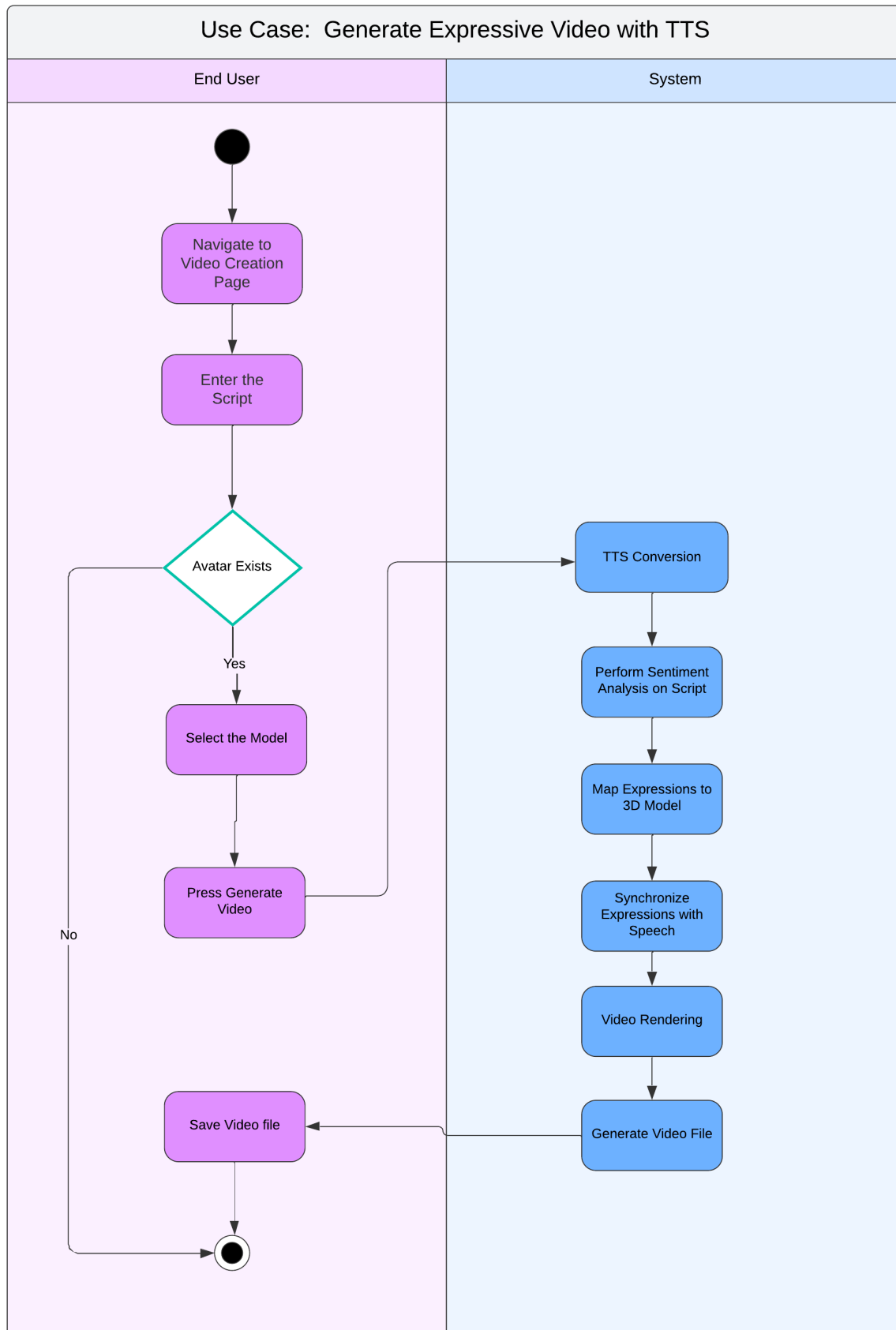


Figure 3.4: Activity Diagram: Generate Expressive Video with TTS

Use Case 4: Generate and Share Content

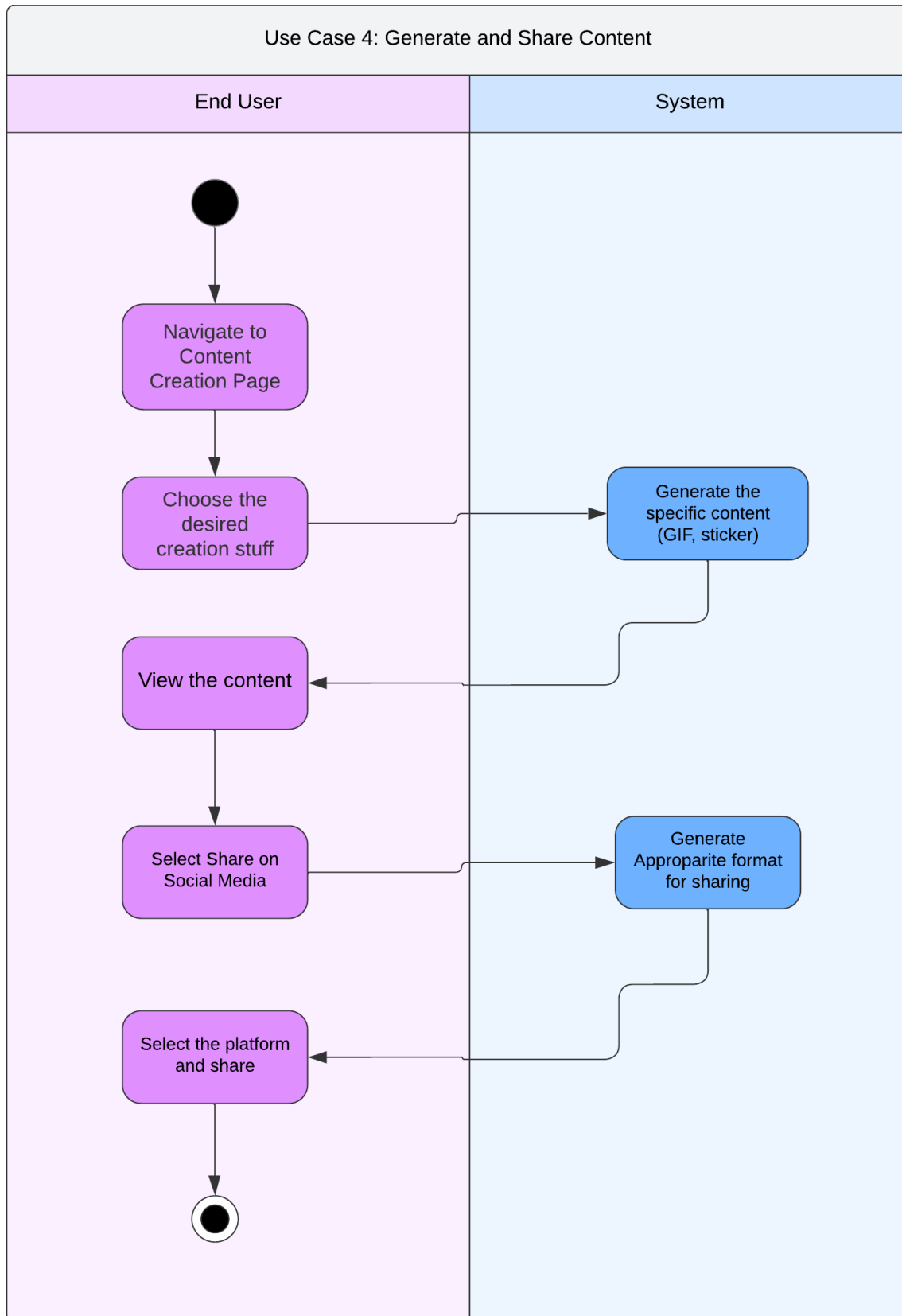


Figure 3.5: Use Case 4: Generate and Share Content

Use Case 5: User Registration

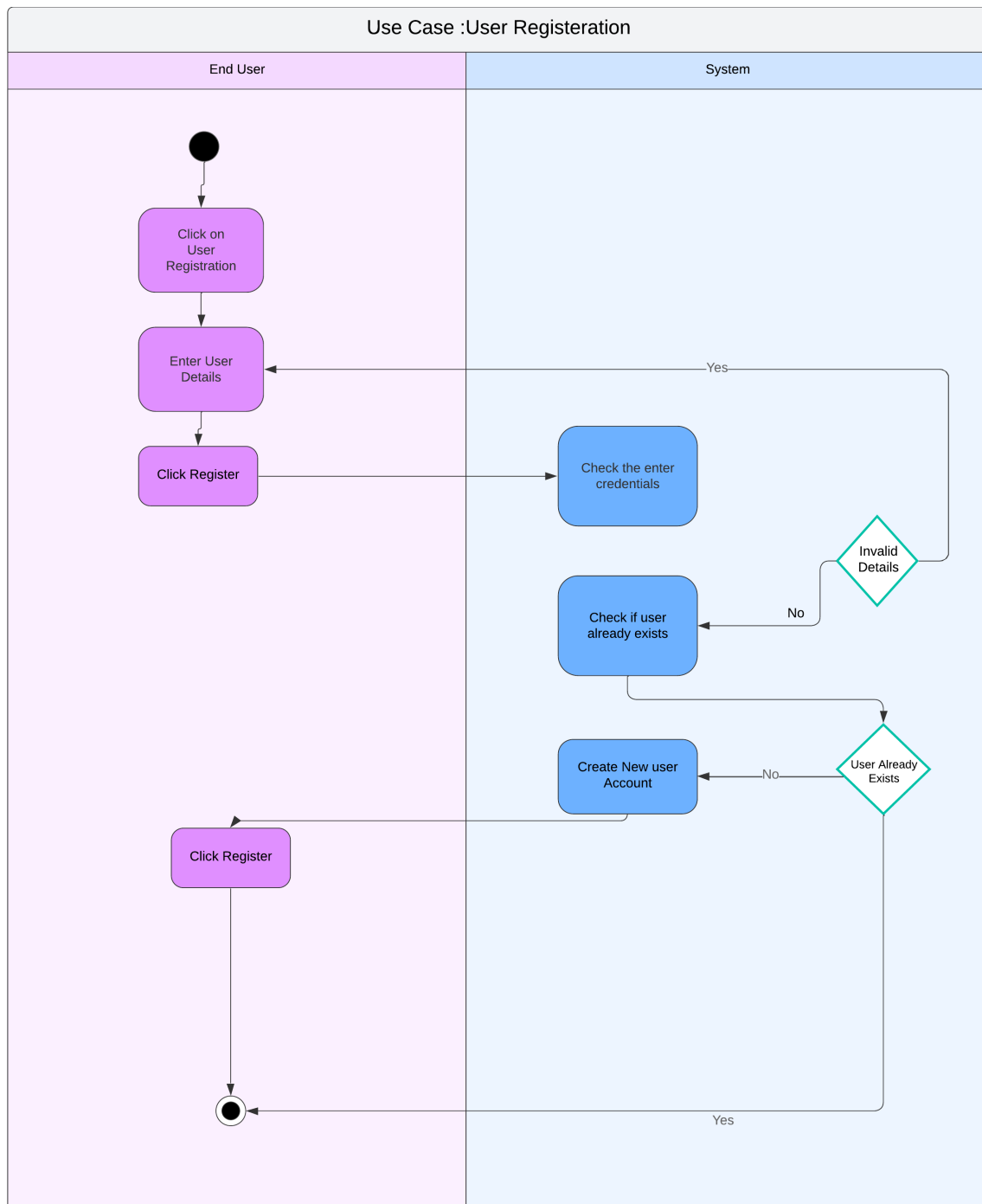


Figure 3.6: Activity Diagram: User Registration

3.2.2 Class Diagram

3.2.2.1 Diagram:

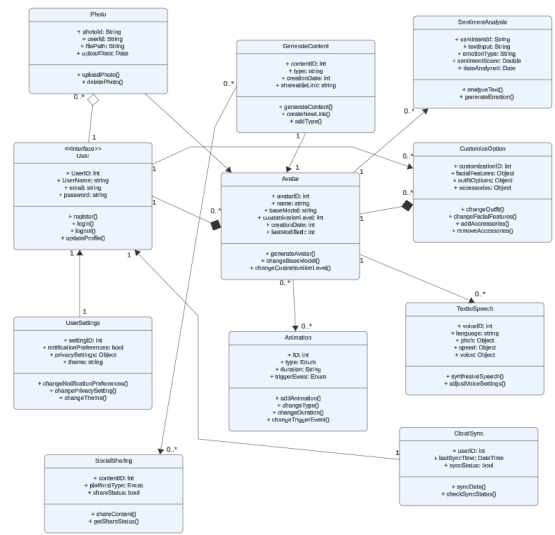


Figure 3.7: Enter Caption

3.2.3 Sequence Diagram

The sequence diagrams capture the interaction between various components and classes involved in key functionalities of the system. They illustrate the sequence of method calls, data exchanges, and flow of control within the application for different use cases. The following sequence diagrams represent crucial aspects of the user journey and system behavior.

3.2.3.1 Use Case: User Registration

- **Description:** This diagram models the steps involved when a user registers on the platform. The process involves user input submission, backend validation, database interaction for storing user details, and confirmation response sent back to the user.
- **Steps:**
 1. User enters registration details.
 2. System validates the input.
 3. System checks if the user already exists in the database.
 4. If valid, the system stores user information in the database and sends a confirmation.

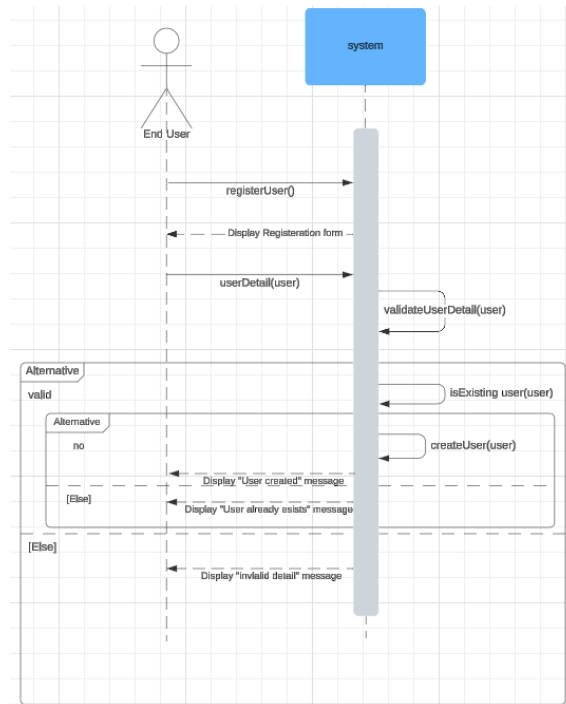


Figure 3.8: User Registration

3.2.3.2 Use Case: Avatar Creation

- **Description:** This sequence diagram showcases the flow where a user uploads an image to create a 3D avatar. The system processes the image, generates the avatar model, and prepares it for further customization.

- **Steps:**

1. User selects or uploads images.
2. System preprocesses the image to ensure it's valid for avatar creation.
3. The backend generates a 3D model mesh of the avatar.
4. The avatar model is stored and prepared for future customizations.

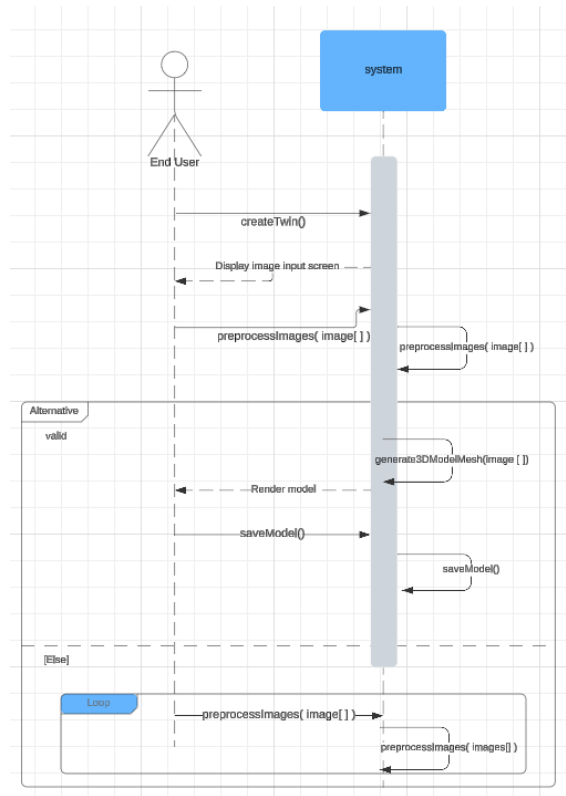


Figure 3.9: Create Avatar

3.2.3.3 Use Case: Avatar Customization

- **Description:** After creating the avatar, the user can customize its features. This diagram demonstrates the interaction between the user interface and backend when the user customizes the avatar's appearance, expressions, and other attributes.

- **Steps:**

1. User selects avatar customization options.
2. System applies the selected customizations (e.g., facial features, expressions).
3. Changes are processed and stored in the system.

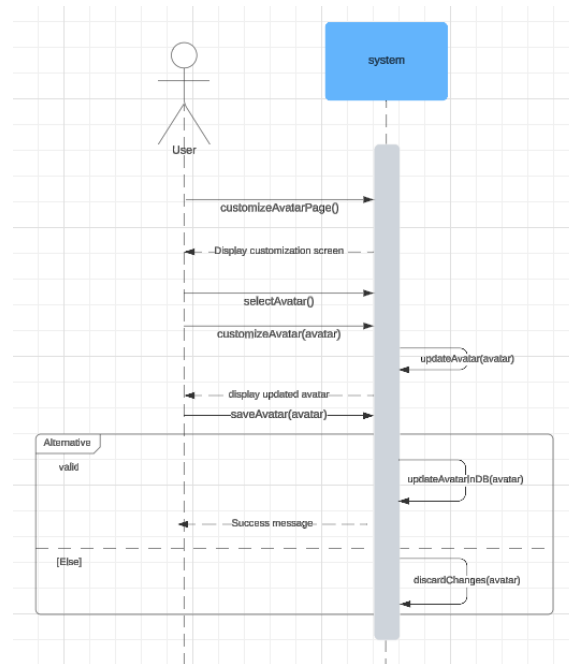


Figure 3.10: Avatar Customization

3.2.3.4 Use Case: Generate Expressive Video with TTS

- **Description:** In this flow, the user selects an avatar and provides a text script to generate an expressive video. The system applies text-to-speech (TTS) conversion, analyzes sentiment, and maps facial expressions to the avatar. The final result is rendered and available for export.

- **Steps:**

1. User selects an avatar and enters a script for video creation.
2. The system converts the text to speech.
3. Sentiment analysis is performed on the text to determine the emotions.
4. The system maps expressions on the avatar to reflect the sentiment of the script.
5. The video is rendered and saved.

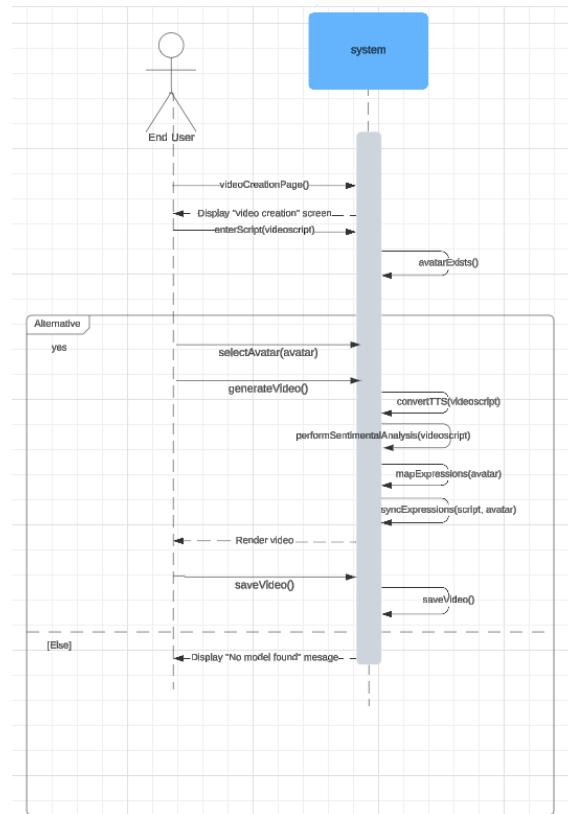


Figure 3.11: Generate Expressive Video with TTS

3.2.3.5 Use Case: Generate and Share Content

- **Description:** This diagram illustrates the process by which the user selects the content format and shares it across various platforms. The system generates the appropriate content format (e.g., GIFs, stickers), displays sharing options, and allows the user to distribute the content.

- **Steps:**

1. User selects the content creation page.
2. System generates the selected type of content.
3. User chooses the format to share the content.
4. The system prepares the content in the desired format and shares it.

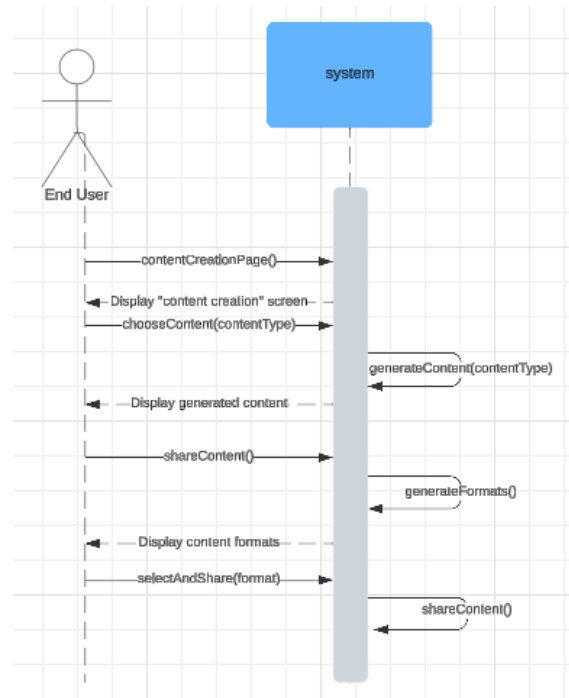


Figure 3.12: Generate and share Content

3.2.4 State Transition Diagram

3.2.4.1 Description:

The state transition diagram models the lifecycle of various entities in the system, particularly focusing on how objects transition between different states based on events or user actions.

- **Example Entities:** User Avatar, Image, text.

- **Possible States:**

- Image Processed
- 3D model created
- Text to speech converted
- Speech analysed
- Avatar ready
- Avatar exported

3.2.4.2 Diagram:

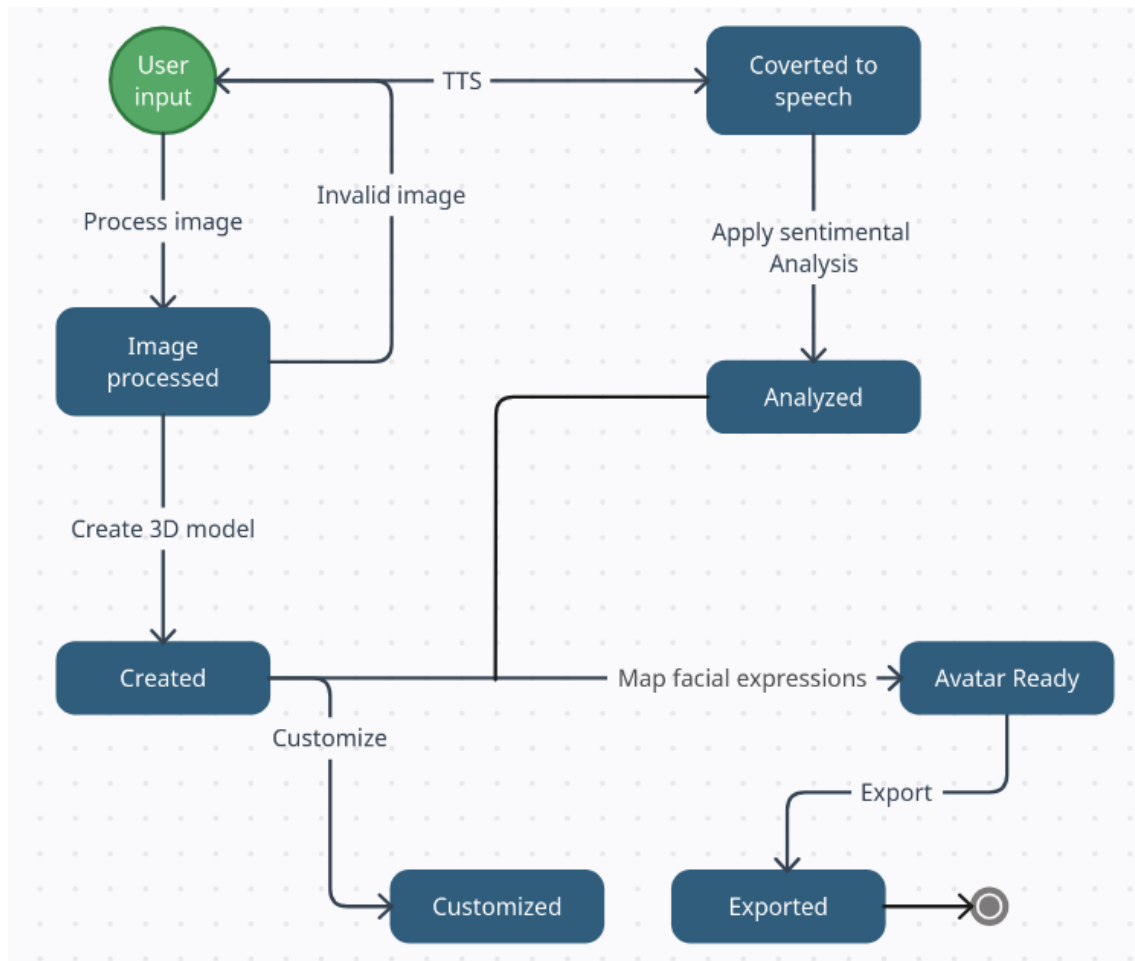


Figure 3.13: state machine diagram

3.3 Data Design

In the ERD of our system, the information domain is represented by entities such as "User," "Avatar," "3DModel," "MediaContent," and their associated attributes like "userID," "avatarID," "modelID," etc. These entities map directly to data structures in MongoDB collections, where each entity is stored as a document.

Major entities such as "User" and "Avatar" contain essential attributes to capture user details and avatar characteristics. MongoDB's flexibility allows for dynamic and hierarchical data organization, making it suitable for managing complex data like 3D models and media content. Data is processed through CRUD operations, where data creation, updating, and retrieval are facilitated by the MongoDB database and structured using

JSON-like documents, maintaining a fluid, schema-less structure ideal for your system's evolving needs.

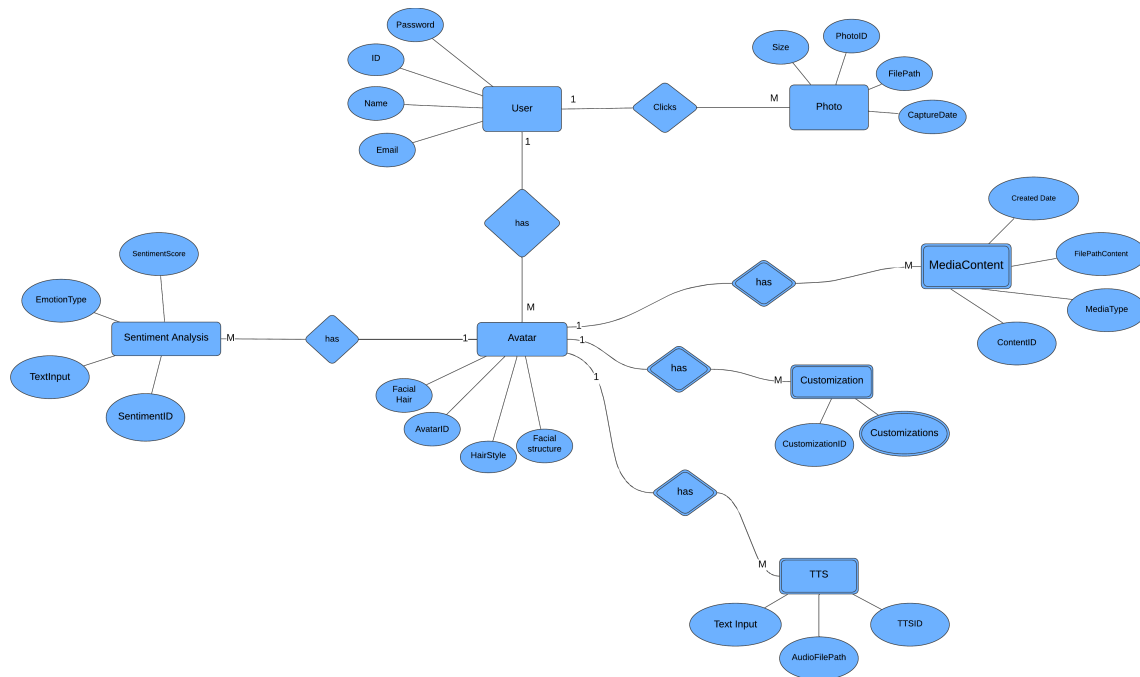


Figure 3.14: ERD-Diagram

Bibliography

- [1] B. Fasel and J. Luetttin. Automatic facial expression analysis: a survey. *Pattern Recognition*, 36(1):259–275, 2003. Accessed: 2024-09-13.
- [2] Y. Furukawa and C. Hernandez. Multi-view stereo: A tutorial. *Foundations and Trends in Computer Graphics and Vision*, 9(1–2):1–148, 2013.
- [3] F. Iqbal, J.M. Hashmi, B.C. Fung, R. Batool, A.M. Khattak, S. Aleem, et al. A hybrid framework for sentiment analysis using genetic algorithm based feature reduction. *IEEE Access*, 7:14637–14652, 2019.
- [4] Liisa Rätsep. Multispeaker text-to-speech synthesis with transformers, 2020. Accessed: 2024-09-13.